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REGULAR ARTICLE

SES-Achievement Gaps in East Asia: Evidence from PISA 2003–2018

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Abstract Widening achievement gaps driven by socioeconomic disparity have become a global concern. Yet, few studies have been able to track the changes and developments of socioeconomic-achievement gaps (SES-achievement gaps henceforward) across time. Using PISA data from 2003 to 2018, we estimated family SES-achievement gaps in seven Asian education systems. The findings suggest that in mainland China the gaps are most pronounced, whereas in Macao, they remain consistently modest. Other systems fall in between these two extremes. Based on the new synchronic and diachronic pieces of evidence, we extrapolate possible explanations of varying SES-achievement gaps across education systems. We found that the gaps tend to be smaller when the low-socioeconomic students perform better in academic environments. Based on our analysis, we argue that if the policy goal is to reduce the learning gap associated with social and economic division, public attention in education should be directed towards students from disadvantaged backgrounds.

Keywords SES · Achievement gap · Educational stratification · PISA

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Introduction

A modern view perceives equality in education as a dissociation between academic achievement and socioeconomic characteristics. In other words, the differences in student achievement should be the result of fair procedures, bereft of the influence of family background (Brighouse & Swift, 2014). Therefore, the impact of socioeconomic status on a child's developmental capacity and academic effort should be removed from the design of social structures such as schooling. Within a fair school system, all children would be adequately equipped with basic knowledge and skills to prepare them for the next step in their academic and professional lives. In this sense, the equality of outcome is measured with metrics such as literacy, numeracy, and comprehension of the natural sciences and humanities (Gosepath, 2014). However, previous studies have reported vigorous effects of socioeconomic factors on children's cognitive development (e.g., Brooks-Gunn & Duncan 1997; Bradley & Corwyn, 2002). Moreover, there are convincing empirical results showing a rising disparity in income and other socioeconomic indicators within almost all countries in the past two decades (Milanovic, 2007; Piketty, 2014). A natural corollary thus propels us to ask, what is the nature of the achievement gap associated with socioeconomic status, and how does it change over time?

To date, efforts to examine the nature of this gap have mushroomed in Western societies. In the past decade, a series of studies have documented robust socioeconomic achievement gaps (Bodovski et al., 2020; Chmielewski, 2019; Linberg et al., 2019). In these studies, the socioeconomic achievement gap (SES-achievement gap, henceforward) is regarded as a new measure for educational inequality, which captures the difference in actual

academic achievement between the most advantaged and disadvantaged groups in the distribution of family socioeconomic status. This approach is of significance in both practical and substantive dimensions. On the practical side, it provides an intuitive look at the extent of educational inequality by generating a straightforward parameter that is easy to examine and analyze among different compositions (e.g., student groups with distinct traits). On the substantive side, the concept of the achievement gap is at the center of contemporary education policy debates. Educational achievement is increasingly seen as an indicator of individual prospects. However, the distribution of educational achievement is far from uniform. The key concern is not a child's innate ability but how the opportunity to learn is tied to family characteristics. Various policy interventions have tried to address the unequal learning opportunities that drive the disparity of learning outcomes. One contributing factor to unequal learning opportunities is family socioeconomic status. It has become the most widely recognized variable since the publication of the Coleman Report in 1966. Rather than specific features within the context of schooling, socioeconomic factors bear a strong relation to academic achievement. Low-SES children are at a disadvantage in developing cognitive skills even before they enter first grade, and this deficiency becomes progressively more remarkable as they progress through higher grade levels. The innovative work by sociologist Sean Reardon (2011) was the first to provide solid retrospective evidence on the income-achievement gaps in the United States. More cross-country comparisons have since been conducted using the context of a family's socioeconomic standing (Chmielewski & Reardon, 2016).

The current study follows suit by concentrating on SES-achievement gaps in East Asian societies. The seven education systems included are Japan, South Korea, mainland China, Hong Kong SAR, Macao SAR, Taiwan, and Singapore. We chose this narrow band of systems for two reasons. First, although research on these gaps has proliferated in Western societies, studies regarding these issues remain rudimentary and incomplete for this part of the world. Second, East Asian education systems share a heavy emphasis on a child's personal and individual effort and believe that all students can learn and achieve through hard work (Stevenson & Stigler, 1992). Among these societies, standardized examinations are commonly viewed as the only opportunity for upward mobility (Cheng, 1998). Indeed, results of the latest international large-scale assessments show that East Asian students typically outperform their counterparts from the rest of the world. They are thus universally acknowledged as global high performers (Mullis et al., 2016; Mullis et al. 2017; OECD 2019b). By singularly focusing on these high-performing systems, we aim to circumvent the standard critique of comparing apples to oranges.[1] Admittedly, there is a long-standing line of research that tries to unpack the mechanisms through which socioeconomic factors lead to differential educational outcomes in the existing literature. However, the temporal focus of how SES-achievement gaps change over time has only emerged more recently. The primary purpose of this study is to investigate the developmental trends of SES-achievement gaps in East Asia. We also made attempts to explore possible explanations of variations in the strengths of the gaps. Our research questions are as follows:

1. How large are the SES-achievement gaps in East Asia?
2. How have these gaps developed and changed in the preceding decade?

First, we reviewed prior literature on SES-achievement gaps, including both synchronic studies and trend studies of such gaps. Next, we describe the details of the quantitative methods used in this study. We then reported the estimated gap sizes and their temporal trends in East Asia based on PISA data from 2003 to 2018. We specifically examined the role of leveling up the performance of the most disadvantaged students in ameliorating the gaps. Finally, we end with a discussion regarding our findings and a final conclusion.

Literature Review

Shifting the Focus of Educational Stratification Research

The powerful influence of family background on student learning performance is well-documented (Brooks-Gunn & Duncan, 1997; Yeung et al., 2002). Under the prevailing family investment model, the practice of child-rearing is regarded as an investment with expectancy. Well-off families are willing to make financial commitments on the cultivation of their offspring, whereas the indigent who struggle to make ends meet cannot afford the equivalent spending. Income is a crucial factor in determining family expenditure on education. For instance, intellectual stimulation is more accessible to affluent families than low-income families (Bradley & Corwyn, 2002). Moreover, affluent parents actively choose homes located in favored school districts (Lareau & Goyette, 2014). They are also more likely to choose better-performing schools within a

[1] Yes, all social science comparisons are comparing apples to oranges at some level. When choosing education systems for comparisons, there is no way to exhaust all the logical possibilities (Martin, 2017, pp. 175–78). In this paper, we want to limit the potential policy tradeoffs due to the "culture" argument (e.g. Meyer & Baker 1996), therefore we are seeing from an Asian perspective.

given range of alternatives (Zhou et al., 2015). With more access to cognitively stimulating materials and advantageous experiences, as a consequence, well-off children tend to have better academic performance than their counterparts.

The education production function framework is commonly used for understanding the varied degrees to which these characteristics matter to learning outcomes across countries. Under this framework, the output of the educational process (academic achievement) is directly related to both family and school inputs (Hanushek, 2020). The former is determined by socioeconomic demographics and is less controllable, whereas the latter are malleable to policy decisions. Based on six decades of comparative studies, the effect of social characteristics on learning is generally weaker in less-developed than in developed countries. Farrell (2013) provides a tentative explanation for this pattern. Since students in developed countries are already abundantly equipped with educational resources, even a modest additional gain in learning level would require difficult and costly inputs. On the contrary, for students in developing countries, even a modest improvement in school quality has great potential of promoting learning progress. This value-added effect is particularly helpful for the most disadvantaged students. Carnoy (1992) emphasizes the active role of state in shaping the education system. Family background has a weaker relation with learning performance where the state takes public responsibility for and is capable of delivering universal high-quality education to students (Carnoy et al., 2007).

However, whether such a productivity pattern holds across different education systems and over time remains less clear. It is only recently that social scientists begin to examine the development of parental socioeconomic characteristics (Reardon, 2011) or to compare the strength of SES in explaining achievement variation between time periods (Byun & Kim, 2010).

Based on the new synchronic and diachronic pieces of evidence (Chmielewski, 2019; Chmielewski & Reardon, 2016; Hanushek et al., 2020), we extrapolate possible explanations of varying SES-achievement gaps. In contrast to economists' mechanical specifications, sociologists have put parents' agency at the center of the social stratification literature—accordingly, a conceivable explanation for varying gaps in the changing social and economic disparity. The advent of "schooled society" has intensified parental emphasis on cognitive development (Baker 2020). High-SES families now have not only more financial resources to invest in intellectual stimulation but also greater access to socioeconomic resources that can benefit their children (Reardon, 2011). In other words, students from high-SES backgrounds enjoy an increasing academic privilege that is beyond the reach of their deprived counterparts.

Trend studies are an essential addition to educational literature. Yet few empirical studies can provide time series of the magnitude of SES-achievement gaps. Even less is a combination of a temporal and a large-scale cross-sectional view. This study aims to fill this void by focusing on East Asia.

Empirical Evidence

Typically, an achievement gap is defined as a difference in average outcomes between two groups (Snyder et al., 2019). This traditional indicator has yielded ample empirical findings.

In the United States, prevailing and persistent socioeconomic achievement gaps have been documented. Students who lived in poverty lag significantly behind their peers in learning performance (Lee & Bowen, 2006). Academic success positively correlates with family income, and an increase in lifetime income corresponds to an advance in achievement (Owens, 2018). Sizable gaps are also present in the United Kingdom, where the SES-achievement gap is much larger than the ethnicity gap and gender gap (Strand, 2014). On a broader scale, a wide variation occurs not only in the size of the SES-achievement gap but also in its connection with average national performance across countries (Akiba et al., 2007). In general, widening SES-achievement gaps are more common in decentralized education systems as well as those with a decreasing investment in education (Broer et al., 2019).

Directly comparing the difference between the mean scores of two groups in socioeconomic distribution gives us a snapshot of these gaps, but such an approach does have its limitations.[2] One glaring issue is that most researchers failed to take variation across time and space into consideration, which in turn confounds the true gap. Secondly, applying the group-mean-difference approach to different test designs ignores the varying measurement errors associated with each type of test, which will inflate the variance of the test score distribution. Thirdly, the group-mean-difference approach masks heterogeneity among the students with a "similar" SES background. Because the distribution of socioeconomic status is different across systems, using standard deviation (SD) metrics to delineate comparison groups can not provide valid comparisons. Achievement gaps derived from this traditional method are, on account of the restrictions, neither desirable to trace their developmental trend across time nor to make comparisons across education systems.

[2] For a detailed discussion on technical and measurement issues, see Reardon (2011), Appendix.

Table 1 Description of education systems

Education system	Sample size						Reading (2018)		Mathematics (2018)		Science (2018)		GDP per capita in PPP dollars (2018)	GINI coefficient (most recent)
	2003	2006	2009	2012	2015	2018	Mean	sd	Mean	sd	Mean	sd		
China (mainland)	–	–	5115	5177	9841	12058	555	87	591	80	590	83	9580	0.47
Singapore	–	–	5283	5546	6115	6676	549	109	569	94	551	97	64579	0.35
Macao	1250	4760	5952	5335	4476	3775	525	92	558	81	544	83	81728	0.36
Hong Kong	4478	4645	4837	4670	5359	6037	524	99	551	94	517	86	48451	0.47
South Korea	5444	5176	4989	5033	5581	6650	514	102	526	100	519	98	33320	0.35
Japan	4707	5952	6088	6351	6647	6109	504	97	527	86	529	92	39304	0.31
Taiwan	–	8815	5831	6046	7708	7243	503	102	531	100	516	99	25008	0.34

GDP data is obtained from International Monetary Fund; Gini coefficients are obtained from these sources

China 2018: 2019 China Yearbook of Household Survey (Department of Household Surveys, National Bureau of Statistics of China)

Singapore 2018: Yearbook of Statistics Singapore, 2019 (Department Of Statistics, Ministry of Trade & Industry, Republic of Singapore)

Macao 2018: Household Budget Survey 2017/2018 (Macao Statistics and Census Service)

Hong Kong 2016: 2016 Population By-census—Thematic Report: Household Income Distribution in Hong Kong (Census and Statistics Department Hong Kong SAR)

South Korea 2018: 2019 Survey of Household Finances and Living Conditions (Statistics Korea)

Japan 2016: 2017 Report on the Survey of Redistribution of Income (Ministry of Health, Labour and Welfare, Government of Japan)

Taiwan 2018: Report on the Survey of Family Income and Expenditure, 2018 (Directorate-General of Budget, Accounting and Statistics Executive Yuan, Republic of China)

In light of such limitations, Reardon (2011) developed an innovative computational approach to estimate the SES-achievement gap by focusing on groups of students who fall on specific distributional locations on the SES spectrum. His measure of SES-achievement gaps ensures that the gaps are comparable across studies, not only those that are conducted in different years but also with studies that collect information in varied ways. This study utilized a similar strategy (Cimentada, 2017). Details are described in the next section.

Data and Method

Data

We use data from the *Programme in International Student Assessment* (PISA), dating from 2003 to 2018 (6 waves in total, with more than 2 million students sampled). PISA belongs to a family of international large-scale assessments conducted by the *Organisation for Economic Co-operation and Development* (OECD). PISA is carried out every three years among OECD countries and partner countries to evaluate 15-year-old students' ability to use their knowledge and skills to meet real-life challenges (OECD, 2017). Additionally, a parent survey is sent to parents to collect family background information. We chose seven education systems from East Asia and listed their background information and sample size in Table 1. The sample size is sufficiently large. According to each wave's technical report, the response rate is also very high.[3]

Measurement

The key variable is the measure of socioeconomic status. In PISA, OECD created such a variable called "*index of economic, social, and cultural status* (ESCS)." It is a well-developed measure that captures three aspects of social status: family possession, parental education, and parental occupation. Moreover, this variable is constructed consistently through each wave of PISA surveys (e.g., OECD, 2017).

In addition to the SES variable, we also utilized the PISA assessment data. PISA tests are conducted in three domains: mathematics, reading, and science, and each test domain is constructed by a panel of international experts (OECD, 2019a). For this study, we used the overall test scores compiled by PISA, not the sub-domain assessment data. Within each cycle, the PISA test data is standardized, making it easy to compare across waves.

Following Reardon (2011), we computed SES-achievement gaps for various groups of students. The 90/10 SES-

[3] The only exception is Macao in 2003, which is the first time that Macao participated in PISA. Due to Macao's small population (< 500000 at that time), the first sample size was 1250. Since then, Macao has counted every 15-year-old in the PISA studies, therefore, one sees a substantial difference in sample size between the first and each consequent PISA cycle.

achievement gap reflects the difference in the average standardized test score of children at the 90th and the 10th percentiles in the family socioeconomic distribution for each participating society within each assessment cycle (henceforth the "90/10 gap"). Similarly, the 90/50 gap and 50/10 gap are the difference in average scores between students at the 90th and the 50th percentiles and at the 50th and the 10th percentiles in the family socioeconomic distribution, respectively. In brief, the 90/10 gap reveals the academic differences between high- and low-socioeconomic students; the 90/50 gap reveals the academic differences between high- and medium-socioeconomic students, and the 50/10 gap reveals the academic differences between medium- and low-socioeconomic students.

The SES-achievement gap can be estimated following a five-step approach:

Step 1 Standardize test scores: Using the appropriate sampling weights for each system within each assessment cycle, we fit a regression model with age measured in months as a sole predictor to adjust the test scores according to student maturation.

$$\hat{Y}_i = Y_i + \epsilon_i = \alpha + \beta_1 \times AGE_i + \epsilon_i, \epsilon \in N(0, \sigma^2) \tag{1}$$

where $\hat{Y}_i$ is the actual student test score, Y_i is the predicted test score, AGE_i is age measured in months (to adjust for maturation), and ϵ_i is the residual for student i. We then standardize the residuals, achievement score net of age influence, to obtain age-adjusted test scores for each student.

Step 2 Convert ESCS variable to SES percentile rank: The ESCS variable is a continuous variable provided by the PISA team. We convert it to a percentile rank for each system within each cycle. The converted SES percentile rank variable ranges from 0 to 1. A percentile rank of 0.1 represents a student at the 10th percentile in the family socioeconomic distribution, while a rank of 0.9 represents one at the 90th percentile in the distribution.

Step 3 Fit a cubic function: To estimate the association between the standardized test score and SES percentile rank, we fit a cubic function. Suppose f can be approximated by a cubic polynomial function. The measures of student achievement and family SES percentile rank are denoted as Y and θ, respectively

$$\begin{aligned} Y &= f(\theta) + \epsilon = a + b\theta + c\theta^2 + d\theta^3 + \epsilon, E(\epsilon|\theta) \\ &= 0 \end{aligned} \tag{2}$$

Step 4 Calculate the difference in average standardized test scores: The fitted cubic curve is used to estimate the difference in average test scores between students at two different percentiles. The 90/10 gap is computed as:

$$\begin{aligned} \hat{\delta}^{90/10} &= [\hat{Y}|\theta = 0.9] - [\hat{Y}|\theta = 0.1] \\ &= [\hat{a} + \hat{b}(0.9) + \hat{c}(0.81) + \hat{d}(0.729)] \\ &\quad - [\hat{a} + \hat{b}(0.1) + \hat{c}(0.01) + \hat{d}(0.001)] \\ &= 0.8\hat{b} + 0.8\hat{c} + 0.728\hat{d} \end{aligned} \tag{3}$$

Likewise, the 90/50 and 50/10 estimators are:

$$\begin{aligned} \hat{\delta}^{90/50} &= [\hat{Y}|\theta = 0.9] - [\hat{Y}|\theta = 0.5] \\ &= [\hat{a} + \hat{b}(0.9) + \hat{c}(0.81) + \hat{d}(0.729)] \\ &\quad - [\hat{a} + \hat{b}(0.5) + \hat{c}(0.25) + \hat{d}(0.125)] \\ &= 0.4\hat{b} + 0.56\hat{c} + 0.604\hat{d} \end{aligned} \tag{4}$$

and

$$\begin{aligned} \hat{\delta}^{50/10} &= [\hat{Y}|\theta = 0.5] - [\hat{Y}|\theta = 0.1] \\ &= [\hat{a} + \hat{b}(0.5) + \hat{c}(0.25) + \hat{d}(0.125)] \\ &\quad - [\hat{a} + \hat{b}(0.1) + \hat{c}(0.01) + \hat{d}(0.001)] \\ &= 0.4\hat{b} + 0.24\hat{c} + 0.124\hat{d} \end{aligned} \tag{5}$$

Step 5 Estimate standard error for the estimators. The variance of the sum of three variables is given by:

$$\begin{aligned} &Var(aX + bY + cZ) \\ &= a^2 Var(X) + b^2 Var(Y) + c^2 Var(Z) \\ &\quad + 2abCov(X, Y) + 2acCov(X, Z) \\ &\quad + 2bcCov(Y, Z) \end{aligned} \tag{6}$$

The standard error of each of these gap estimators is computed from the estimated variance–covariance matrix of the regression as following, using the 90/10 estimator as an example:

$$\begin{aligned} se(\hat{\delta}^{90/10}) &= [Var(0.8\hat{b} + 0.8\hat{c} + 0.728\hat{d})]^{1/2} \\ &= [0.64Var(\hat{b}) + 0.64Var(\hat{c}) \\ &\quad + 0.53Var(\hat{d}) + 1.28Cov(\hat{b}, \hat{c}) \\ &\quad + 1.165Cov(\hat{b}, \hat{d}) + 1.165Cov(\hat{c}, \hat{d})]^{1/2} \end{aligned} \tag{7}$$

Likewise, the standard errors for 90/50 and 50/10 estimators could be derived similarly.

Combining SES-Achievement Estimator With Plausible Values

Due to the unique design of PISA, the performance level of each student was estimated with a set of plausible values (usually five or ten). To compute point estimates of the SES-achievement gap using PISA achievement data, one simply needs to average the sets of estimates from each plausible value (OECD, 2009b). However, plausible values introduce additional variations in the estimation due to complex sampling design; thus, it is necessary to explicitly take such variability in the computation of standard errors into account. Secondary analyses with plausible values are conducted according to Rubin (1987)'s rules.

We repeat the same five-step analysis for all the plausible values. The estimated results (5 for PISA 2003–2012 and 10 for PISA 2015 and 2018) are combined into an arithmetic average. Since the average is an estimate, we also consider the uncertainty arising from the estimation process. Thus, the standard error $\sqrt{T}$ is computed, which follows a Student's t-distribution. A detailed calculation of the standard error is included in Appendix B.

Results

Sizes of the SES-Achievement Gaps in 2018

In Table 2, we present various SES-achievement gap estimators together with their standard errors for each education system using the latest PISA 2018 data. It is evident that low-SES students fall behind their high-SES counterparts in educational achievement, as all the 90/10 estimates are positive numbers in standard metrics across three test domains. Overall, most systems in our analysis show 90/10 gaps between 0.5 and 1 SD. In particular, Macao presents the smallest of the 90/10 gaps (they were consistently smaller than 0.5 SD) regardless of test domain, which was in strong contrast with mainland China, which shows the largest 90/10 gap (significantly larger than 1 SD).

Specifically, in terms of mathematics performance, the 90/10 gap varies from 0.47 to 1.12 SD. All education systems show a gap above 0.5 SD except Macao. Singapore, Taiwan, and mainland China's achievement gaps are above 1 SD. For reading, the 90/10 gaps range between 0.37 and 1.25 SD. Following a similar pattern, Macao is the only education system whose gap is smaller than 0.5 SD, whereas both Singapore and mainland China have a gap larger than 1 SD. By the same token, in science, Macao presents the smallest gap of 0.35 SD while mainland China

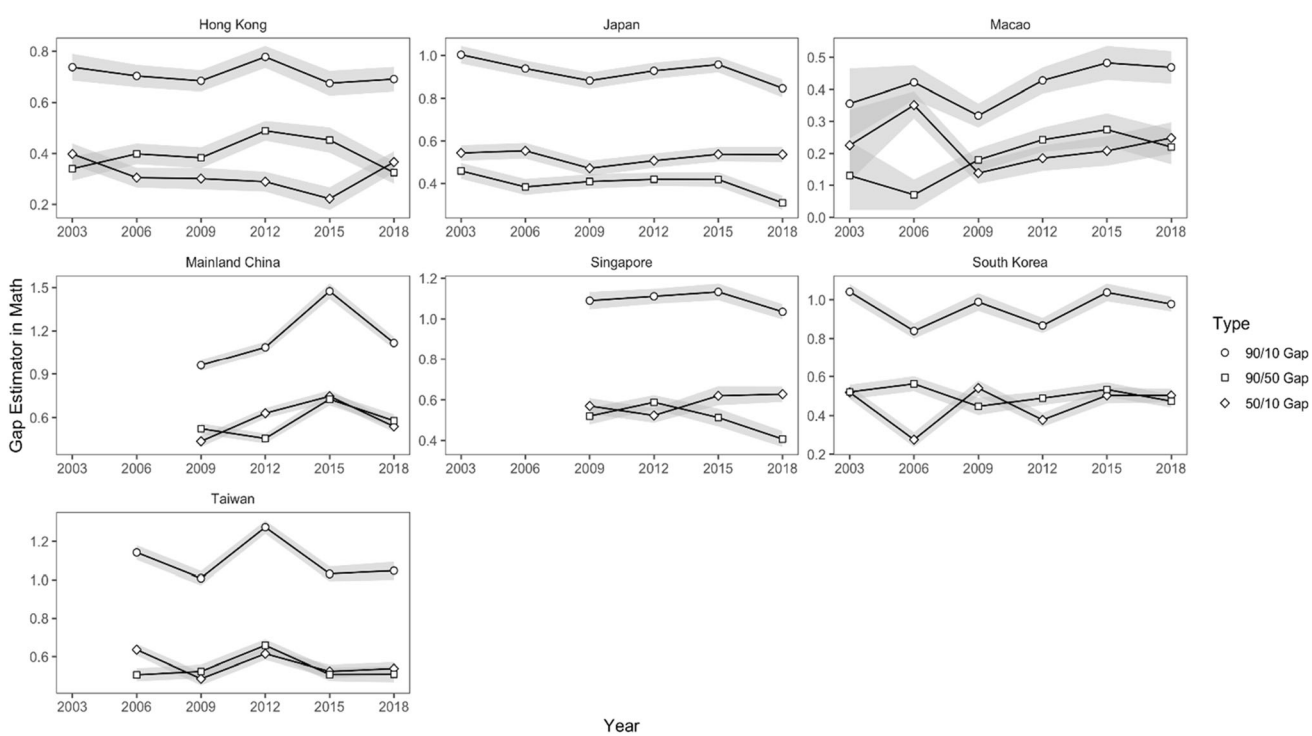

Fig. 1 SES-achievement gap trends in Math

Table 2 SES-achievement gap estimation using PISA 2018

Education system	Gap	Reading		Mathematics		Science	
		Estimate	SE	Estimate	SE	Estimate	SE
China (mainland)	90/10	1.25	0.04	1.12	0.04	1.15	0.05
	90/50	0.60	0.04	0.58	0.04	0.58	0.05
	50/10	0.65	0.04	0.54	0.04	0.57	0.04
Singapore	90/10	1.02	0.03	1.04	0.04	1.05	0.04
	90/50	0.42	0.03	0.41	0.04	0.41	0.03
	50/10	0.59	0.03	0.63	0.04	0.64	0.03
Macao	90/10	0.37	0.05	0.47	0.05	0.35	0.05
	90/50	0.20	0.05	0.22	0.05	0.22	0.05
	50/10	0.17	0.04	0.25	0.05	0.13	0.05
Hong Kong	90/10	0.69	0.04	0.69	0.05	0.72	0.04
	90/50	0.33	0.04	0.32	0.04	0.44	0.04
	50/10	0.37	0.04	0.37	0.04	0.29	0.04
South Korea	90/10	0.83	0.04	0.98	0.04	0.82	0.04
	90/50	0.36	0.03	0.47	0.03	0.38	0.04
	50/10	0.47	0.03	0.50	0.03	0.44	0.04
Japan	90/10	0.80	0.04	0.85	0.04	0.78	0.04
	90/50	0.30	0.03	0.31	0.03	0.30	0.03
	50/10	0.49	0.03	0.54	0.04	0.48	0.04
Taiwan	90/10	0.99	0.03	1.05	0.05	0.99	0.04
	90/50	0.52	0.03	0.51	0.04	0.51	0.04
	50/10	0.47	0.03	0.54	0.03	0.48	0.04

has the largest at 1.15 SD. Most education systems manifest a science gap between 0.7 and 1 SD.

We repeat the same analysis to estimate the 90/50 and 50/10 gaps. As a standard inference in linear regression, one would expect that SES and test scores are positively correlated. Therefore it is not a surprise that the 90/50 gaps are not as large as the 90/10 gaps. In general, the 90/50 gaps lie between 0.2 and 0.6 SD. The 90/50 gaps are generally below 0.5 SD with the exception of Taiwan and mainland China. Once again, Macao maintains the smallest 90/50 gap, while mainland China has the largest.

The 50/10 gaps are smaller than the 90/10 gaps but generally greater than the 90/50 gaps. This finding shows attenuating performance differentials as students move up the SES ladder. Overall, the 50/10 gaps fall between 0.13 and 0.65 SD, indicating a slightly larger range than the 90/50 gaps. Relatively large 50/10 gaps (i.e., those larger than 0.5 SD) are observed in mainland China and Singapore in all three test domains. Conversely, the 50/10 gaps in Macao remain constantly the smallest.

Evidence from the latest PISA data suggests that there are consistent and considerable SES-achievement gaps in East Asia. Students with favorable socioeconomic backgrounds tend to have better academic performance than their counterparts. Moreover, such gaps become less obvious in the upper SES hierarchy. Additionally, among all the education systems included in this study, Macao holds the smallest SES-achievement gaps. By contrast, the gaps in mainland China are prominently large.

Trends of the SES-Achievement Gaps

In the second section, we present the results by tracking achievement gap estimates for each system across time and overlap these trend lines on a singular graph. The trends for the SES-achievement gap in mathematics are presented in Fig. 1 (reading and science estimates are graphed in Figures 3, 4. They are qualitatively similar). The shaded areas represent the standard error of each gap estimate. Note that the magnitude of SES-achievement gaps are pretty different across systems, reflected on the range of the y-axis. Although the data covered an average timespan of 15 years at three-year intervals, no clear-cut direction of movement was observed in SES-achievement gaps; instead, the gaps rose and fell in most education systems. It should be noted that the estimators in the current study stand for academic achievement gaps between students of different positions in the SES distribution within an education system in any given year. The underlying meaning of SES at any given time or location could and may very well change.

However, here the focus is on the difference between two *relative* positions in the SES distribution rather than absolute changes in SES. That is to say, the 90th percentile in any given SES distribution indexed different students across time and locations. However, our focus is on the learning outcomes *between* two specific groups of students. In the competition for educational opportunities, a socially scarce good, it is not how much one has but if others own more than one that makes a difference (Hirsch, 1976). The gap estimates reflect the disparity of learning outcomes associated with the positional difference of students.

Overall, SES-achievement gaps did not increase or decrease at a consistent rate for any single education system. Instead, they tended to fluctuate over time. These enduring gaps suggest that it would be almost impossible to close them in the short term. The 90/10 gaps are consistently larger than the 90/50 and 50/10 gaps and display more variation over time than the other two estimates.

In terms of system-specific findings, the 90/10 gaps fluctuated turbulently in mainland China, Taiwan, and South Korea in comparison to those in Macao, Hong Kong, Japan, and Singapore. Mainland China held the most fluctuating 90/10 gaps, with the largest gap of 1.47 SD in 2015, in strong contrast to 1.09 and 1.12 SD in the previous and following assessment cycles. This is likely due to the varying sampling design used in different cycles. Japan seemed to narrow the 90/10 gaps with the latest 2018 gap estimate of 0.85 versus 1 SD using the earliest data in 2003. Despite its most recent 90/10 gap being slightly larger than that initially, Macao has always maintained the smallest 90/10 gaps.

The diachronic trend of the 90/50 gap reveals some similar tendencies as those in the 90/10 gaps. Again, mainland China exhibited the most varying degrees of 90/50 gaps. Meanwhile, both Japan and Singapore experienced a slight reduction in 90/50 gaps; their latest 90/50 gaps were 0.15 and 0.11 SDs smaller than their beginning estimates. The 90/50 gaps in Macao, in spite of a minor upward tendency, remained on the low side. Similarly, we can observe variations in 50/10 gaps across time in mainland China and South Korea; other education systems retained the magnitude without an obvious upward or downward tendency. We also plotted trends in reading and science and found similar results. For simplicity, these charts are included in the Appendix.

To summarize, there is no clear tendency for any single education system given the development of gaps as a consequence of many fluctuations across. The magnitude of the gaps did not experience significant change for each system during the last decade.

Comparison of Academic Achievement Among Low-SES Students

The previous two sections present a synchronic (RQ1) and diachronic (RQ2) view of SES-achievement gaps. Nevertheless, what could policymakers do to close such gaps effectively? In this section, we offer a preliminary analysis. Intuitively, large gap estimates could be driven by the high performance of high-SES students (i.e., heightened SES-achievement gradient at the right tail of the SES distribution). It could also be driven by the low performance of low-SES students. However, from an education policy perspective, the second scenario has undoubtedly caught more attention as attenuating the SES-achievement gradients for the students from the left tails of the SES distribution has been central to several policy interventions in the past (e.g., Glewwe et al., 2017; Carnoy et al., 2007). In this section, we stepped further to understand whether such practices exist in our samples.

We estimated students' average scores at the five bottom deciles of the socioeconomic distributions across these education systems in all three test domains based on data from all available PISA cycles. Figure 2 presents the estimated math scores derived from the latest 2018 assessment (Estimates from all three subjects are listed in Tables 3, 4, 5). Overall, as students moved up their respective socioeconomic ladders, their corresponding performance levels also increased. This finding fits with the conventional wisdom that socioeconomic status is positively correlated with academic achievement. In addition, this analysis also reveals that the achievement gap between two adjacent lower tiers in the SES distribution was more extensive than that between two adjacent upper tiers. For instance, the difference between the 10th and 20th percentile was larger than the difference between the 20th and 30th percentile. In addition, the gradient of academic performance was steeper at the bottom and then progressed to becoming smaller towards the top. That is to say, overall, across different education systems, the lower the students are placed on the socioeconomic ladder, the *more* they lagged behind, even compared to their peers who were also considered "low-SES."

This analysis sketched an explanandum of SES-achievement gaps, which broadens our view into typically reported "mean performance." All the educational systems include in our analysis are regarded as high-performing, yet they are subject to different internal dynamics. Macao has the smallest SES-achievement gradient (i.e., Macao's curve in Fig. 2 is the flattest). It was most significant in the lowest category, the 10th percentile, and gradually shrunk towards the upper end of the SES distribution. Low socioeconomic students from Hong Kong scored the second-highest in the bottom three deciles. This partially explained the findings

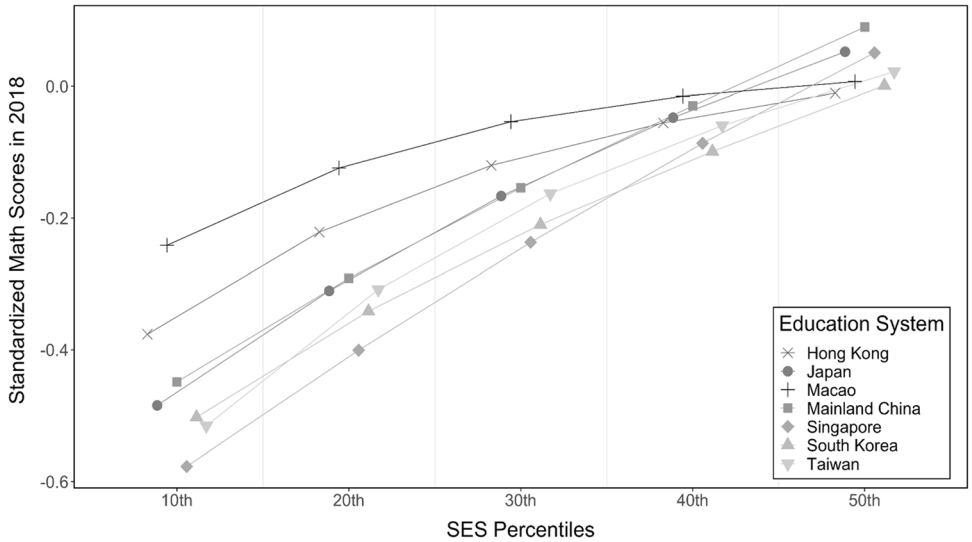

Fig. 2 Estimated math scores of low-SES students in PISA 2018

from the previous section that Macao and Hong Kong maintained smaller SES-achievement gaps among these education systems.

This finding could guide future studies and policy research in understanding the nature of achievement gaps. For instance, both Singapore and mainland China displayed sizable 90/10 gaps in mathematics (Fig. 1). But students at the 10th percentile of the SES distribution in Singapore scored much lower than their peers in mainland China (Fig. 2). We corroborated Fig. 2 with point estimates in Tables 3, 4, 5. We find that China's SES-achievement gaps could be driven by an extremely high performance from high-SES students coupled with a normal performance from low-SES students; Singapore's gap could be driven by a lower performance from low-SES students and a normal performance from high-SES students.[4]

This depiction of the internal dynamics of each system opens up new research possibilities beyond our current space. For instance, researchers can look more specifically at practices and policies that drive the performance of specific student groups while inhibiting others. What could be done to restrict the SES-achievement gaps? One possibility certainly lies in boosting the performance of low-SES students.

[4] There is no doubt the language we used here is relative. This comparison is not norm-referenced. Terms such as "high performance" or "normal performance" are only indicative in the comparative context here. We do not wish to leave the impression that Singapore's low-SES students performed "not well" in an absolute sense.

Discussion

The current study provides a diachronic view of SES-achievement gaps in East Asia based on the latest international assessment data. Useful insights can be drawn from the Asian experience, especially from Macao as a successful gap minimizer.

The magnitude of our estimated SES-achievement gaps in East Asia is close to income-achievement gaps among OECD countries, which is on average about one standard deviation (Chmielewski & Reardon, 2016). Our analysis reaffirms that educational stratification is a shared issue for not only world developed economies but also Asian societies. Based on data from PISA 2006, we found comparable 90/50 gaps in South Korea with those generated by Chmielewski and Reardon (2016). However, our 50/10 gaps are significantly larger than theirs. Our finding suggests that the students at the bottom SES distribution who lacked wealth, parental education, and prestigious occupation were negative impacted than poverty alone could have done. Moreover, we observed a fluctuating SES-achievement gap in mainland China, whose 90/10 math gap ranges from 0.96 SD in 2009 to 1.47 SD in 2015. We argue that this might be caused by inconsistent sampling approaches across PISA assessment cycles. In 2009 and 2012, Shanghai represented mainland China, then in 2015, Beijing, Jiangsu, and Guangdong joined the assessment, and in 2018 Zhejiang replaced Guangdong. The extremely large gaps in 2015 could partly be explained by a large proportion of the low-SES population in Guangdong province. (And this regular PISA top-performer encountered a

much lower ranking in a national average performance in 2015.) Regardless, mainland China has the largest SES-achievement gaps across the board.

Macao has maintained narrow SES-achievement gaps and the average performance of 15-year-olds is much better than that of their counterparts from most areas of the world. Because Macao included every 15-year-old in the PISA study, the estimates are the most accurate. Though a preliminary analysis, we suggest that such a low SES-achievement gap could be attributed to the fact that disadvantaged students from Macao outperformed other Asian disadvantaged students in our analysis. This comes as no surprise, as Macao embraces an inclusionary educational approach, creating universal access to schooling.[5] The special administrative region is the first region to make fifteen-year free education universally available in Greater China, covering from early childhood to senior secondary education (Lau, 2017). Over 90 percent of local schools are tuition-free. The rest are low-cost since students receive tuition subsidies from the government. In addition to the normal course of study, government-funded extra-curricular activities are also free to all school children. It is common for schools to offer voluntary supplementary instruction for vulnerable students. Moreover, financial assistance for school meals, study materials and auxiliary equipment is easily accessible to students from families with economic difficulties. On a broader scale, the government has heavily invested in public services, and a wide range of cultural and recreational facilities are open to the public, such as libraries, sports stadiums, and study rooms. In a nutshell, the government has managed to take affirmative actions to secure shared, and more importantly, sufficient opportunities to learn.

While the stereotypical image of testing culture is pervasive in East Asian societies, this study reveals variations and dynamics among different systems, echoing the varying effects of SES on academic achievement documented in Marks and Pokropek (2019). For example, a standard polemic is to compare Hong Kong and Singapore (La Londe and Verger 2020; Lee & Gopinathan, 2018). Yet Singapore's SES-achievement gaps are significantly more extensive than those of Hong Kong. One plausible explanation might be due to the lopsided educational design in Singapore, which favors those who have already demonstrated excellent academic performance and are considered to be more suitable for the academic track. Taiwan, which has eased the pressure of testing (Hwang, 2015; Yang, 2004), actually maintained quite a high level of SES-achievement gaps. Meanwhile, albeit public-funded learning programs have been carried out for years (Cheng and Jacob, 2016), low-SES students from Taiwan still fell behind their counterparts from most Asian societies in this study. This could be due to insufficient policy concerns to low-SES students: most programs are aimed at cultural minorities rather than disadvantaged students (Sung et al., 2014). These findings all suggest that there might be undisclosed factors that drive the SES-achievement gaps. This paper should be interpreted as inviting more fine-grained future research on the policy and practices of different systems, a topic which we could only scratch the surface here.

In sum, overall excellence in international achievement tests should not be taken as the sole goal of educational development. All the Asian systems included in this study are regarded as "high-performing," yet some show persistently high levels of SES-achievement gaps. In fact, theorists such as David Labaree (1997) have long pointed out that as education has become an increasingly prominent private good due to its prevailing influence in promoting social mobility, the role of education as a public good is sharply undermined especially in the realization of democratic equality. Excessive competition for social efficiency and social mobility might result in severe stratification in education, as manifested by large socioeconomic achievement gaps. Although this study focuses on SES-achievement gaps in East Asia, our findings have policy implications for almost every education system around the globe.

Limitation

This current study is the first to investigate the SES-achievement gaps in East Asia. It is built on emerging sociology of education techniques. Nevertheless, limited by the data, we could not fully explain the changes in gaps across time. The current study is descriptive and a first step into the internal dynamics of student performances. Readers should bear in mind that PISA data does not include many context-specific variables, such as reform initiatives or social policy environments, for further modeling. Perhaps researchers could investigate the causal mechanism behind SES-achievement gaps in the future by combining multiple data sources.

A second limitation of this study lies in the trend analysis. Unlike studies from western societies (Chmielewski, 2019; Reardon, 2011), data points within a single educational system across time included in this study are limited. For most systems, we have four-to-six data points, as such, is hard to determine the movement of the socioeconomic achievement gap with certainty. Readers should not simply look at the data points at the start and end of the year, for

[5] A more detailed outline of Macao's macro education policy can be read here: https://portal.dsej.gov.mo/webdsejspace/internet/Inter_main_page.jsp?id=21211.

both starting and ending years in the study are somewhat arbitrary. If more accumulated data points are included (i.e. alternative assessment and SES measures), then the trend of gaps will be more robust.

The diachronic trend of the socioeconomic achievement gaps in East Asia described in this study provides empirical evidence for determining educational equality. To further confirm these results, it is helpful to conduct replication studies based on other international datasets. Future research should focus on the tendency of SES-achievement gaps covering a longer time span.

Appendix

Appendix A. SES-Achievement Gap Trends in Reading and Science

See Figs. 3, 4.

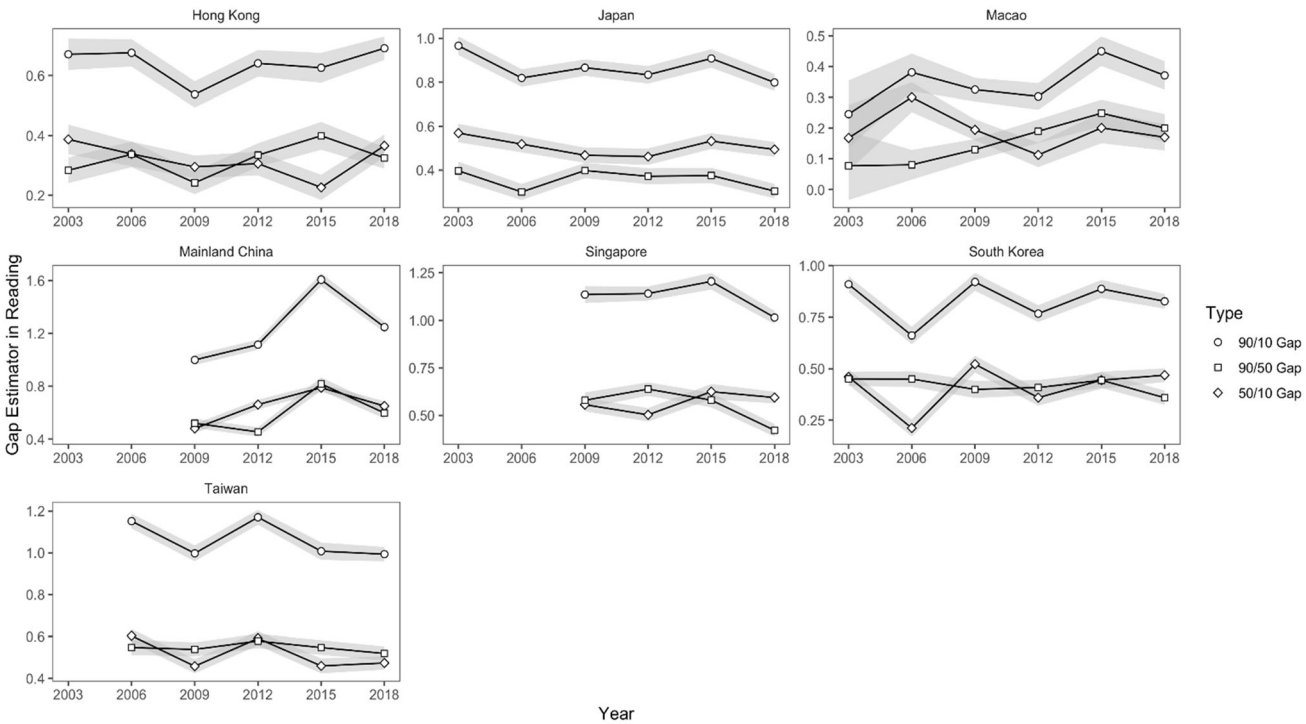

Fig. 3 SES-achievement gap trends in reading

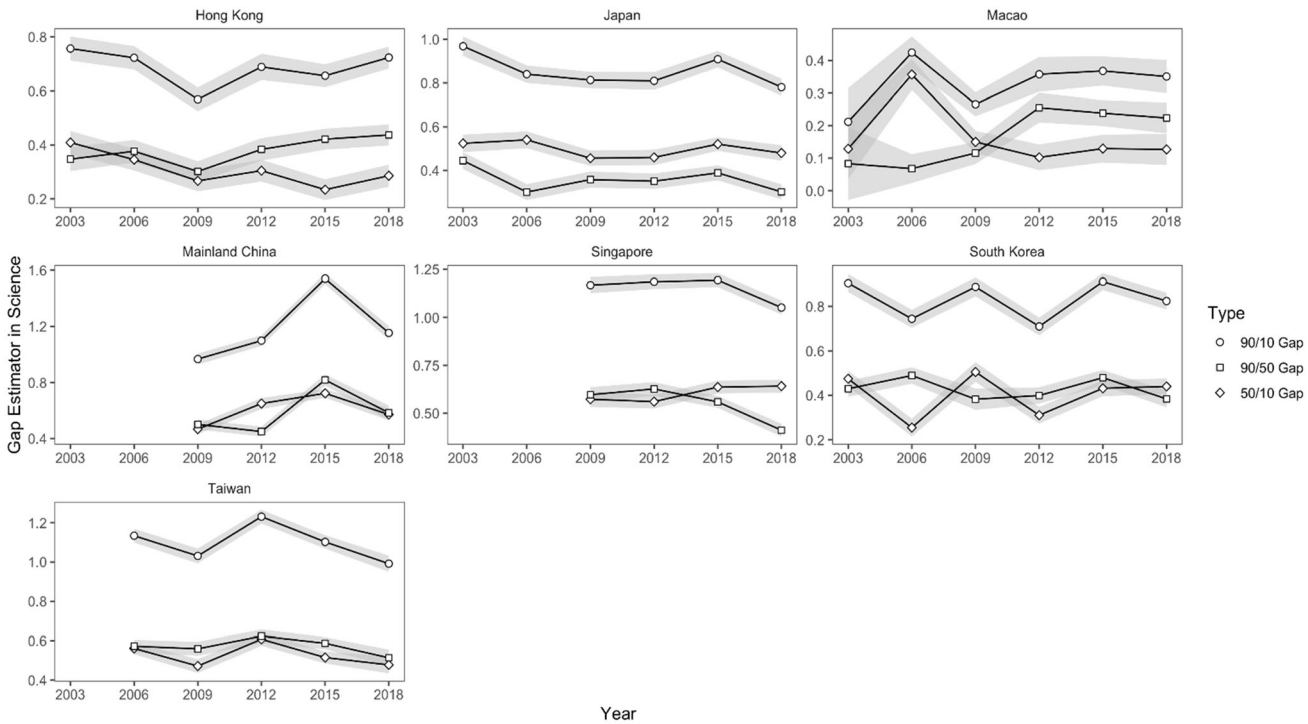

Fig. 4 SES-achievement gap trends in science

Appendix B. Computing SES-Achievement Gaps and Respective Standard Errors

Given PISA uses an assessment design known as the multiple-matrix sampling (Shoemaker, 1973), none of the students completed the full-length test but rather were allocated a portion of the test (Rutkowski et al., 2010). As a result, although the traditional "average out" approach still works for the point estimate, it does not work for the standard errors. Therefore, secondary analysis with plausible values follows Rubin (1987)'s rules.

Let $\overline{Q}$ denote the final point estimate and U its estimated variance. Here are equally plausible estimates $\hat{Q}_1,\hat{Q}_2,\ldots,\hat{Q}_m$ (equals to 5 or 10 in PISA) and their corresponding variances $U_1,U_2,\ldots,U_m$.

We first compute the overall estimate, namely an average of all point estimates, which is given by

$$\overline{Q} = \frac{1}{m}\sum_{i=1}^{m}\hat{Q}_i.$$

The within-imputation variance is simply the average of the estimated variances:

$$\overline{U} = \frac{1}{m}\sum_{i=1}^{m}U_i.$$

The between-imputation variance is the sample variance of the estimates themselves:

$$B = \frac{1}{m-1}\sum_{i=1}^{m}(\hat{Q}-\overline{Q})^2.$$

The total variance, is the sum of variance within and between imputation with an additional correction factor to account for the simulation error in $\overline{Q}$,

$$T = \overline{U} + \left(1+\frac{1}{m}\right)B$$

The square root of T is the overall standard error associated with $\overline{Q}$. The 95% confidence interval can be obtained using the approximation

$$\overline{Q} \pm t_{df}\sqrt{T},$$

where t_{df} denotes a quantile of Student's t-distribution with degrees of freedom

$$df = (m-1)(1+\frac{m\overline{U}}{(m+1)B})^2.$$

P-values can be obtained by comparing the ratio $\overline{Q}/\sqrt{T}$ to the same t-distribution (Schafer & Olsen, 1998).

Appendix C. Mean Performance of PISA Scores at Specific SES Percentiles

See Tables 3, 4, 5.

Table 3 Mean performance of PISA score at specific SES percentiles (Math)

Region	Year	Mean performance at specific SES percentiles								
		10th	20th	30th	40th	50th	60th	70th	80th	90th
Hong Kong	2003	− 0.42	− 0.24	− 0.12	− 0.06	− 0.03	0.00	0.05	0.15	0.31
		(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)	(0.03)	(0.04)
	2006	− 0.36	− 0.23	− 0.15	− 0.10	− 0.06	− 0.02	0.05	0.16	0.34
		(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
	2009	− 0.33	− 0.23	− 0.15	− 0.09	− 0.03	0.03	0.11	0.21	0.35
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)
	2012	− 0.34	− 0.25	− 0.18	− 0.12	− 0.05	0.02	0.12	0.26	0.44
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
	2015	− 0.31	− 0.21	− 0.15	− 0.11	− 0.09	− 0.04	0.03	0.16	0.37
		(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)
	2018	− 0.38	− 0.22	− 0.12	− 0.06	− 0.01	0.03	0.09	0.18	0.31
		(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
Japan	2003	− 0.52	− 0.31	− 0.16	− 0.06	0.02	0.09	0.18	0.30	0.48
		(0.03)	(0.02)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)	(0.03)
	2006	− 0.53	− 0.34	− 0.19	− 0.07	0.02	0.11	0.20	0.29	0.41
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2009	− 0.47	− 0.33	− 0.20	− 0.09	0.01	0.10	0.20	0.30	0.42
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.49	− 0.32	− 0.18	− 0.08	0.02	0.10	0.19	0.30	0.44
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2015	− 0.51	− 0.31	− 0.16	− 0.05	0.03	0.10	0.18	0.29	0.45
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2018	− 0.48	− 0.31	− 0.17	− 0.05	0.05	0.14	0.21	0.29	0.36
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
Macao	2003	− 0.20	− 0.15	− 0.09	− 0.03	0.02	0.07	0.11	0.14	0.16
		(0.08)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.07)	(0.06)	(0.08)
	2006	− 0.28	− 0.14	− 0.04	0.03	0.07	0.10	0.12	0.13	0.14
		(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)	(0.03)	(0.04)
	2009	− 0.15	− 0.07	− 0.03	− 0.01	− 0.01	− 0.01	0.01	0.07	0.17
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.20	− 0.09	− 0.03	− 0.01	− 0.02	− 0.01	0.01	0.09	0.23
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)	(0.03)
	2015	− 0.23	− 0.11	− 0.05	− 0.03	− 0.02	− 0.01	0.03	0.11	0.26
		(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)
	2018	− 0.24	− 0.12	− 0.05	− 0.01	0.01	0.03	0.06	0.12	0.23
		(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)
China (mainland)	2009	− 0.46	− 0.31	− 0.19	− 0.10	− 0.03	0.06	0.16	0.30	0.50
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.59	− 0.34	− 0.17	− 0.05	0.04	0.11	0.20	0.32	0.50
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2015	− 0.64	− 0.32	− 0.12	0.01	0.11	0.20	0.32	0.52	0.83
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.04)
	2018	− 0.45	− 0.29	− 0.15	− 0.03	0.09	0.21	0.34	0.49	0.67
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)	(0.04)

Table 3

Singapore	2009	− 0.53	− 0.32	− 0.17	− 0.05	0.04	0.13	0.23	0.37	0.56
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.02)	(0.03)
	2012	− 0.52	− 0.37	− 0.23	− 0.11	0.00	0.12	0.25	0.41	0.59
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2015	− 0.58	− 0.38	− 0.22	− 0.08	0.04	0.16	0.27	0.40	0.55
		(0.03)	(0.02)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2018	− 0.58	− 0.40	− 0.24	− 0.09	0.05	0.17	0.28	0.38	0.46
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
South Korea	2003	− 0.52	− 0.29	− 0.14	− 0.06	0.00	0.06	0.14	0.29	0.52
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2006	− 0.35	− 0.22	− 0.15	− 0.11	− 0.08	− 0.03	0.07	0.23	0.49
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
	2009	− 0.54	− 0.32	− 0.17	− 0.07	0.00	0.06	0.14	0.26	0.44
		(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)
	2012	− 0.41	− 0.29	− 0.19	− 0.11	− 0.03	0.05	0.16	0.29	0.46
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2015	− 0.51	− 0.31	− 0.18	− 0.08	− 0.01	0.07	0.17	0.31	0.53
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2018	− 0.50	− 0.34	− 0.21	− 0.10	0.00	0.10	0.20	0.33	0.48
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
Taiwan	2006	− 0.64	− 0.37	− 0.19	− 0.08	0.00	0.07	0.16	0.30	0.51
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2009	− 0.50	− 0.32	− 0.20	− 0.10	− 0.01	0.07	0.18	0.32	0.51
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.61	− 0.36	− 0.20	− 0.08	0.01	0.10	0.22	0.40	0.66
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2015	− 0.51	− 0.31	− 0.17	− 0.07	0.01	0.09	0.18	0.32	0.52
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2018	− 0.52	− 0.31	− 0.16	− 0.06	0.02	0.10	0.20	0.34	0.53
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)

Table 4 Mean Performance of PISA score at specific SES percentiles (Reading)

Region	Year	Mean performance at specific SES percentiles								
		10th	20th	30th	40th	50th	60th	70th	80th	90th
Hong Kong	2003	− 0.40	− 0.20	− 0.08	− 0.03	− 0.01	0.00	0.03	0.11	0.27
		(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
	2006	− 0.37	− 0.21	− 0.12	− 0.06	− 0.03	0.01	0.06	0.15	0.31
		(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)
	2009	− 0.29	− 0.20	− 0.13	− 0.06	0.00	0.06	0.11	0.18	0.24
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
	2012	− 0.32	− 0.20	− 0.12	− 0.06	− 0.01	0.04	0.10	0.19	0.33
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)
	2015	− 0.29	− 0.18	− 0.12	− 0.09	− 0.07	− 0.03	0.03	0.15	0.33
		(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
	2018	− 0.38	− 0.19	− 0.08	− 0.03	− 0.01	0.01	0.05	0.15	0.32
		(0.03)	(0.02)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
Japan	2003	− 0.52	− 0.28	− 0.11	− 0.02	0.05	0.10	0.16	0.27	0.44
		(0.03)	(0.02)	(0.03)	(0.03)	(0.02)	(0.02)	(0.03)	(0.02)	(0.03)
	2006	− 0.48	− 0.29	− 0.15	− 0.04	0.04	0.11	0.18	0.25	0.34
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2009	− 0.46	− 0.32	− 0.20	− 0.09	0.01	0.10	0.19	0.29	0.40
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.45	− 0.28	− 0.15	− 0.06	0.02	0.08	0.16	0.26	0.39
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2015	− 0.50	− 0.28	− 0.13	− 0.03	0.03	0.09	0.16	0.26	0.41
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2018	− 0.45	− 0.29	− 0.16	− 0.05	0.04	0.12	0.20	0.27	0.35
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
Macao	2003	− 0.14	− 0.14	− 0.11	− 0.04	0.03	0.09	0.14	0.15	0.11
		(0.08)	(0.06)	(0.07)	(0.06)	(0.06)	(0.06)	(0.07)	(0.06)	(0.08)
	2006	− 0.23	− 0.09	0.00	0.05	0.07	0.08	0.09	0.11	0.15
		(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)
	2009	− 0.18	− 0.09	− 0.03	0.00	0.02	0.03	0.05	0.08	0.15
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.13	− 0.02	0.02	0.01	− 0.02	− 0.05	− 0.04	0.02	0.17
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
	2015	− 0.21	− 0.10	− 0.04	− 0.02	− 0.01	− 0.01	0.02	0.10	0.24
		(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
	2018	− 0.18	− 0.09	− 0.04	− 0.02	− 0.01	0.00	0.03	0.09	0.19
		(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
China (mainland)	2009	− 0.49	− 0.31	− 0.18	− 0.09	− 0.01	0.06	0.16	0.30	0.51
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.62	− 0.36	− 0.18	− 0.05	0.05	0.13	0.22	0.33	0.50
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2015	− 0.69	− 0.35	− 0.14	0.00	0.10	0.20	0.35	0.57	0.92
		(0.02)	(0.02)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.04)
	2018	− 0.54	− 0.32	− 0.15	− 0.01	0.11	0.23	0.36	0.52	0.71
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)

Table 4

Singapore	2009	− 0.54	− 0.33	− 0.18	− 0.07	0.02	0.11	0.22	0.38	0.60
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.52	− 0.35	− 0.22	− 0.11	− 0.02	0.09	0.22	0.39	0.62
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2015	− 0.60	− 0.40	− 0.24	− 0.10	0.02	0.14	0.27	0.42	0.60
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2018	− 0.55	− 0.38	− 0.23	− 0.09	0.04	0.16	0.26	0.36	0.46
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
South Korea	2003	− 0.46	− 0.26	− 0.13	− 0.05	0.00	0.06	0.14	0.26	0.45
		(0.03)	(0.02)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)	(0.03)
	2006	− 0.28	− 0.19	− 0.13	− 0.10	− 0.07	− 0.02	0.06	0.19	0.38
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)
	2009	− 0.52	− 0.29	− 0.14	− 0.05	0.01	0.06	0.12	0.23	0.40
		(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)
	2012	− 0.37	− 0.26	− 0.17	− 0.09	− 0.01	0.06	0.15	0.26	0.39
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)	(0.03)
	2015	− 0.44	− 0.28	− 0.16	− 0.07	0.00	0.08	0.17	0.28	0.44
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2018	− 0.45	− 0.30	− 0.17	− 0.07	0.02	0.10	0.18	0.27	0.38
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
Taiwan	2006	− 0.62	− 0.38	− 0.21	− 0.10	− 0.01	0.07	0.17	0.32	0.53
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2009	− 0.48	− 0.35	− 0.23	− 0.12	− 0.02	0.08	0.20	0.35	0.52
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.57	− 0.33	− 0.17	− 0.06	0.02	0.10	0.20	0.36	0.60
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2015	− 0.48	− 0.32	− 0.20	− 0.11	− 0.02	0.07	0.18	0.33	0.53
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2018	− 0.47	− 0.27	− 0.15	− 0.06	0.00	0.07	0.16	0.31	0.52
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)

Table 5 Mean Performance of PISA score at specific SES percentiles (Science)

Region	Year	Mean performance at specific SES percentiles								
		10th	20th	30th	40th	50th	60th	70th	80th	90th
Hong Kong	2003	− 0.43	− 0.22	− 0.10	− 0.05	− 0.03	− 0.01	0.04	0.14	0.32
		(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
	2006	− 0.39	− 0.23	− 0.14	− 0.08	− 0.04	− 0.01	0.06	0.16	0.33
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)
	2009	− 0.29	− 0.20	− 0.13	− 0.07	− − 0.02	0.03	0.10	0.18	0.28
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
	2012	− 0.33	− 0.23	− 0.15	− 0.09	− 0.02	0.05	0.13	0.23	0.36
		(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)
	2015	− 0.31	− 0.20	− 0.14	− 0.10	− 0.07	− 0.03	0.04	0.16	0.35
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
	2018	− 0.34	− 0.21	− 0.13	− 0.08	− 0.06	− 0.02	0.06	0.18	0.38
		(0.03)	(0.02)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
Japan	2003	− 0.50	− 0.30	− 0.15	− 0.05	0.02	0.09	0.18	0.29	0.47
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)	(0.03)
	2006	− 0.50	− 0.30	− 0.15	− 0.04	0.04	0.11	0.18	0.25	0.35
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2009	− 0.44	− 0.30	− 0.18	− 0.08	0.01	0.10	0.18	0.27	0.37
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.44	− 0.27	− 0.15	− 0.05	0.02	0.09	0.16	0.25	0.37
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2015	− 0.49	− 0.29	− 0.15	− 0.05	0.03	0.09	0.17	0.27	0.42
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2018	− 0.44	− 0.29	− 0.16	− 0.05	0.04	0.12	0.19	0.26	0.34
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
Macao	2003	− 0.12	− 0.09	− 0.05	− 0.02	0.01	0.04	0.07	0.09	0.10
		(0.07)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.07)	(0.07)	(0.08)
	2006	− 0.27	− 0.11	0.00	0.06	0.08	0.09	0.10	0.11	0.15
		(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
	2009	− 0.14	− 0.06	− 0.01	0.01	0.01	0.01	0.02	0.05	0.12
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.14	− 0.04	− 0.01	− 0.02	− 0.04	− 0.05	− 0.03	0.05	0.21
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.04)
	2015	− 0.16	− 0.06	− 0.02	− 0.02	− 0.03	− 0.03	− 0.01	0.06	0.21
		(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)
	2018	− 0.15	− 0.07	− 0.03	− 0.02	− 0.03	− 0.02	0.01	0.07	0.20
		(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)

Table 5

China (mainland)	2009	− 0.48	− 0.30	− 0.18	− 0.09	− 0.01	0.07	0.16	0.30	0.49
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.61	− 0.35	− 0.17	− 0.05	0.04	0.12	0.20	0.32	0.49
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)	(0.03)
	2015	− 0.65	− 0.34	− 0.15	− 0.02	0.08	0.18	0.32	0.55	0.90
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2018	− 0.48	− 0.28	− 0.13	− 0.01	0.10	0.20	0.33	0.48	0.68
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.04)
Singapore	2009	− 0.55	− 0.34	− 0.19	− 0.08	0.02	0.12	0.24	0.40	0.62
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.56	− 0.38	− 0.24	− 0.11	0.00	0.12	0.26	0.42	0.63
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2015	− 0.60	− 0.41	− 0.25	− 0.10	0.03	0.16	0.29	0.43	0.59
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
	2018	− 0.59	− 0.40	− 0.24	− 0.08	0.06	0.18	0.29	0.38	0.47
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
South Korea	2003	− 0.46	− 0.24	− 0.11	− 0.03	0.01	0.05	0.12	0.24	0.44
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2006	− 0.32	− 0.21	− 0.14	− 0.10	− 0.06	− 0.01	0.07	0.21	0.43
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)	(0.03)
	2009	− 0.49	− 0.30	− 0.16	− 0.06	0.01	0.07	0.15	0.25	0.39
		(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
	2012	− 0.34	− 0.24	− 0.16	− 0.09	− 0.03	0.05	0.13	0.24	0.37
		(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
	2015	− 0.44	− 0.28	− 0.17	− 0.08	− 0.01	0.06	0.16	0.29	0.47
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2018	− 0.43	− 0.27	− 0.15	− 0.06	0.01	0.08	0.16	0.26	0.39
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
Taiwan	2006	− 0.59	− 0.36	− 0.21	− 0.11	− 0.03	0.05	0.15	0.31	0.54
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2009	− 0.50	− 0.35	− 0.23	− 0.12	− 0.02	0.08	0.20	0.35	0.53
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2012	− 0.60	− 0.35	− 0.18	− 0.07	0.01	0.10	0.21	0.38	0.64
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
	2015	− 0.53	− 0.34	− 0.20	− 0.10	− 0.02	0.08	0.19	0.35	0.57
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
	2018	− 0.47	− 0.29	− 0.16	− 0.07	0.01	0.09	0.18	0.32	0.52
		(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)

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